

无细胞表达芯片的组成

无细胞
表达系统
(IVT)

哺乳动物
表达体系

兔网织红细胞体系（得率低、修饰有限、速度慢）

CHO裂解体系（得率高、修饰全面、速度慢）

HeLa裂解体系（得率高、修饰全面、速度快）✓

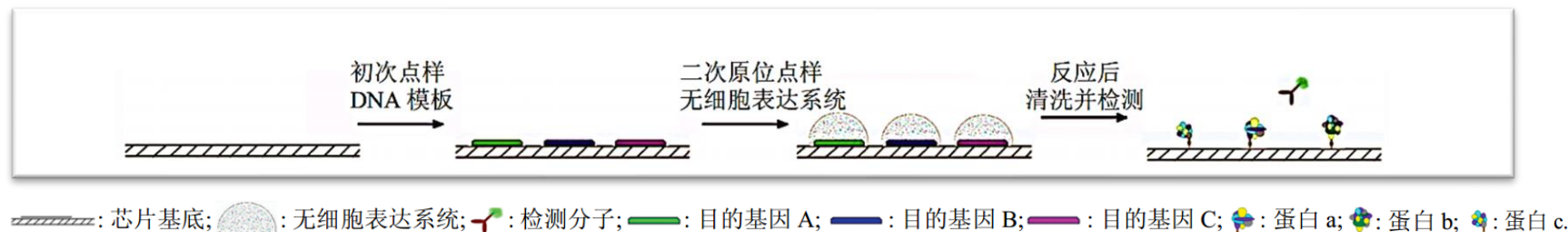
表达载体

pT7CFE1-CHis 载体（C端单标签）

pT7CFE1-CGFP-HA-His 载体（C端多标签）

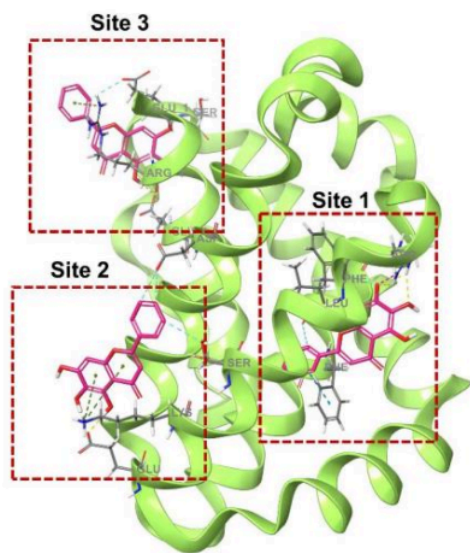
pT7CFE1-CHis-RADA 载体（公司改造，适用于芯片固定）✓

液滴封装技术
(首创)

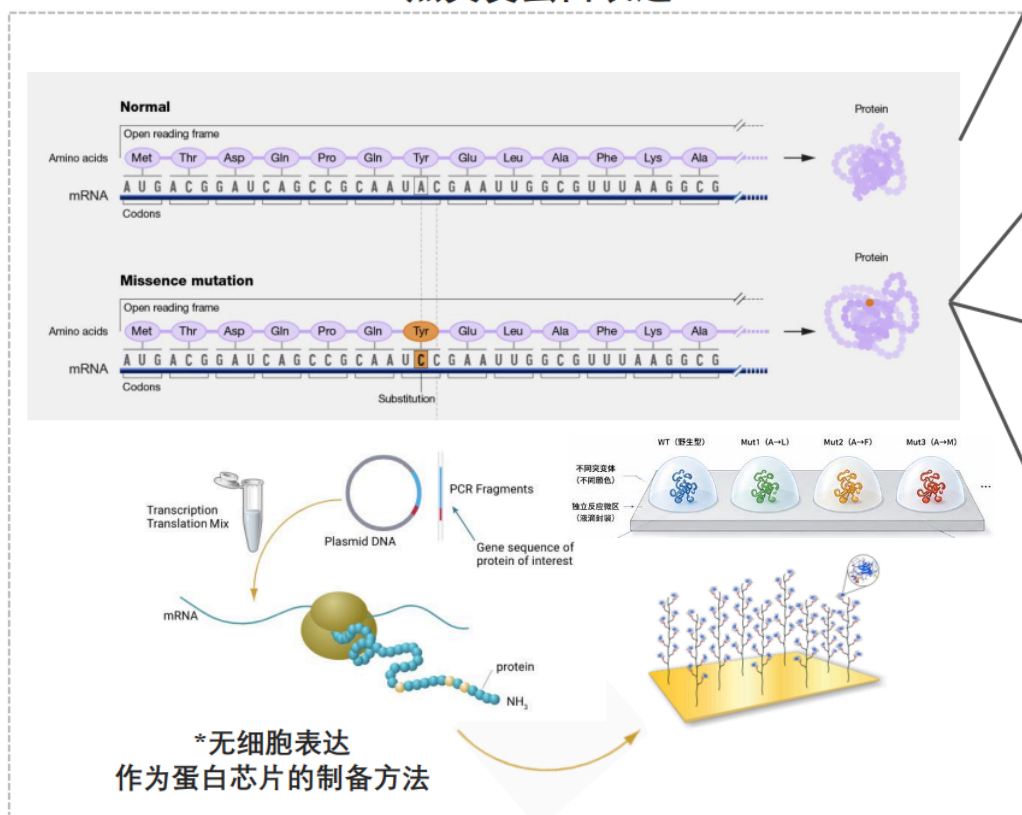


无细胞表达芯片用于点突变研究

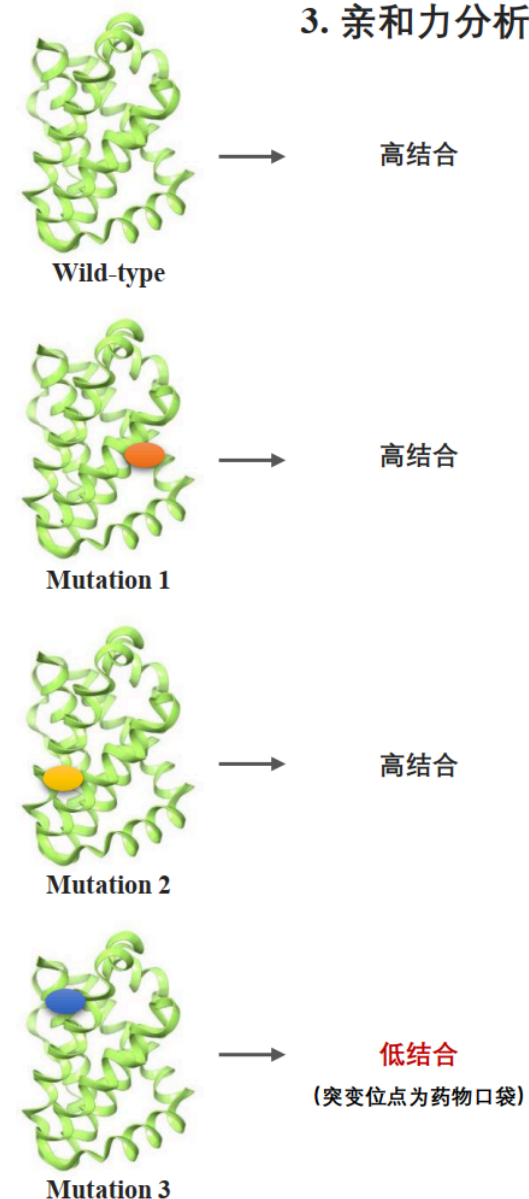
1. 对接预测



2. 点突变蛋白表达



3. 亲和力分析



应用案例 1 —— 多种蛋白并行验证

WT	:	TTC	GGG	AAG	CGC	GAG	GGG	CAC
		F	G	K	R	E	G	H
R39A	:	TTC	GGG	AAG	GCC	GAG	GGG	CAC
		F	G	K	A	E	G	H

WT	:	GAC	GGA	GCC	ACT	TAC	GAA	GGG
		D	G	A	T	Y	E	G
T52A	:	GAC	GGA	GCC	GCC	TAC	GAA	GGG
		D	G	A	A	Y	E	G

WT	:	TAC	ATC	TAC	AGC	GAC	GGC	GGC
		Y	I	Y	S	D	G	G
S168A	:	TAC	ATC	TAC	GCC	GAC	GGC	GGC
		Y	I	Y	A	D	G	G

WT	:	TGG	GCT	CAC	GAC	ATG	AAG	GAC
		W	A	H	D	M	K	D
D203D	:	TGG	GCT	CAC	GCC	ATG	AAG	GAC
		W	A	H	A	M	K	D

WT	:	GCG	AAC	GGC	GAC	CGC	TTC	AAA
		A	N	G	D	R	F	K
D263A	:	GCG	AAC	GGC	GCC	CGC	TTC	AAA
		A	N	G	A	R	F	K

图1 蛋白点突变设计方案

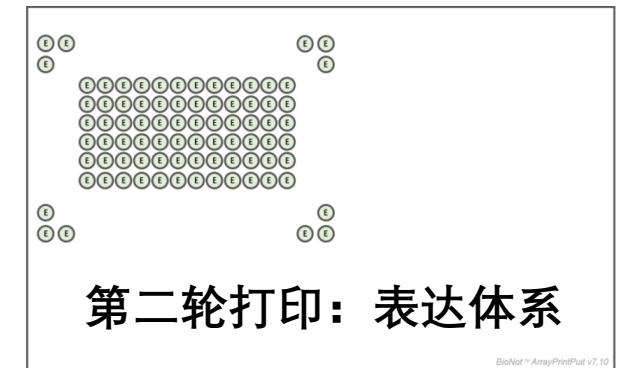
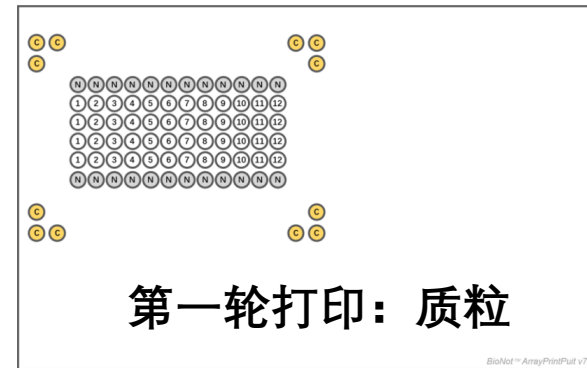


图2 芯片打印流程

应用案例 1 —— 多种蛋白并行验证

图3 蛋白原位表达后芯片表面信号扫描图
(表达后质检)

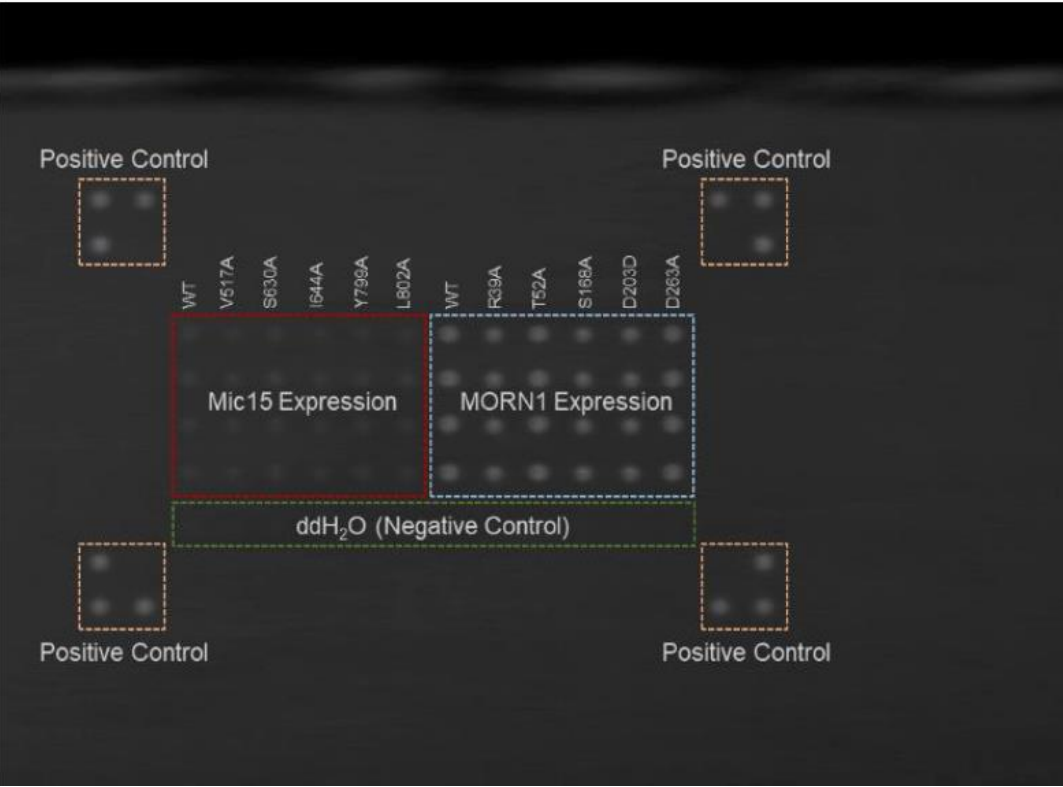


表1 蛋白表达量的标定及统计

分组	名称	分子量 (kDa)	表达时长 (min)	共振信号 (RU)	产物质量 (ng)	产物摩尔量 (fMol)
Positive-1	RefProtein	12.000	360	44650	95.77	7.98
Positive-2	RefProtein	12.000	360	41971	90.02	7.50
Positive-3	RefProtein	12.000	360	40292	86.42	7.20
Positive-4	RefProtein	12.000	360	37471	80.37	6.70
Negative-1	ddH2O	0.018	360	21	0.05	2.50
Negative-2	ddH2O	0.018	360	22	0.05	2.62
Negative-3	ddH2O	0.018	360	23	0.05	2.74
Negative-4	ddH2O	0.018	360	24	0.05	2.86
Mic15	MIC15_WT	168.035	360	31452	67.46	0.40
Mic15	MIC15_Y779A	168.035	360	29250	62.74	0.37
Mic15	MIC15_L802A	168.035	360	31590	67.76	0.40
Mic15	MIC15_I644A	168.035	360	32221	69.11	0.41
Mic15	MIC15_V517A	168.035	360	29321	62.89	0.37
Mic15	MIC15_S630A	168.035	360	32253	69.18	0.41
MORN1	MORN1_WT	40.890	360	49813	106.84	2.61
MORN1	MORN1_D263A	40.890	360	49813	106.84	2.61
MORN1	MORN1_S168A	40.890	360	52303	112.18	2.74
MORN1	MORN1_D203D	40.890	360	52826	113.31	2.77
MORN1	MORN1_R39A	40.890	360	58108	124.64	3.05
MORN1	MORN1_T52A	40.890	360	52297	112.17	2.74

应用案例 1 —— 多种蛋白并行验证

图4 亲和力曲线

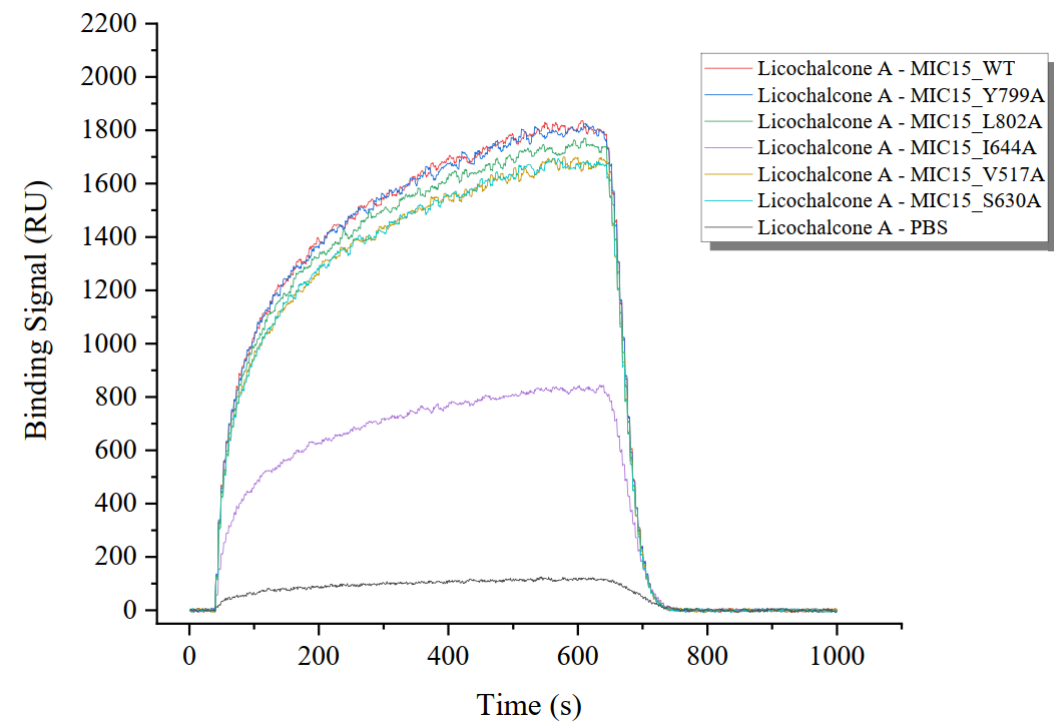
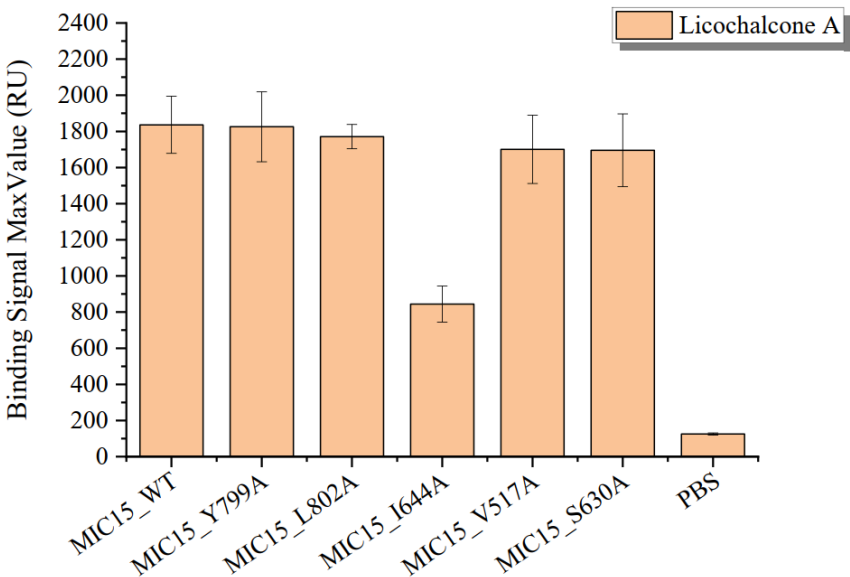


图5 结合信号最大值



No.	固定相 (打印于芯片表面)	流动相 (流经芯片表面)	Avg ka (1/Ms)	Avg kd (1/s)	Avg KD (M)	Int.Intensity Level	ABS (tr_KD)
1	MIC15_WT	Licochalcone A	1.23E+05	1.30E-03	1.06E-08	Strong	26.497
2	MIC15_Y779A	Licochalcone A	1.21E+05	1.33E-03	1.10E-08	Strong	26.438
3	MIC15_L802A	Licochalcone A	7.99E+04	1.49E-03	1.87E-08	Strong	25.672
4	MIC15_I644A	Licochalcone A	2.17E+02	8.66E-02	3.99E-04	Weak	11.290
5	MIC15_V517A	Licochalcone A	5.11E+04	1.28E-03	2.51E-08	Strong	25.248
6	MIC15_S630A	Licochalcone A	5.06E+04	1.06E-03	2.10E-08	Strong	25.505